SARA AI

Smart **A**ccessible **R**eader & **A**ssistant

Agentic AI Study Assistant for Visually Impaired K-12 Students

👨‍💻 Creators & Developers

Aditya Rai & Sparsh Ruhela

🎓 Mentor: **Vineeta Garg** | 🏫 Shaheed Rajpal DAV Public School

✦ **One of the Greatest Innovations of the Decade** ✦

30+
Voice Commands

8
Languages

20+
AI Features

253M
Can Benefit

" Created with love for the ones who can make this world a beautiful place to live in "

THE PROBLEM

253 Million

Visually Impaired People Worldwide

Yet most AI tools are completely unusable without sight.

0	AI tutors built for VI students
1 in 7	people have vision impairment

What visually impaired students face:

Cannot use most study apps

Interfaces rely entirely on visual interaction — buttons, menus, drag & drop

No personalised AI tutor

Generic chatbots give the same answer to everyone regardless of grade or topic

Voice tools are too basic

Existing tools just read text aloud — no teaching, quizzing or adaptive learning

Miss critical exam reminders

No AI tool reads their timetable & sends reminders at the right time

Language & accessibility barriers

Most tools work only in English with no font/spacing controls for dyslexia

OUR SOLUTION

Meet SARA — Not just a chatbot. A complete AI companion.

AI Teacher

4-step structured lessons
for any topic, any grade

Email Reminders

AI reads timetable & sends
real exam alerts via email

Agentic AI

Autonomously plans, tracks
and adapts to each student

8 Languages

Hindi, Tamil, Telugu,
French, Spanish & more

100% Voice

30+ commands — complete
control without a screen

Accessibility

VI mode, Dyslexia mode,
full font & spacing controls

Single HTML file • No server • No installation • Free forever • Works on any browser

SARA AI vs Normal AI Chatbot

Normal AI Chatbot	VS	SARA AI
Just answers questions typed by user		Teaches, plans, analyses & acts autonomously
Forgets everything the moment you close		Tracks all quiz scores, weak topics & progress
Cannot be used without seeing the screen		100% voice controlled — zero screen required
Only produces text — no audio output		Reads every response aloud in chosen language
English only — no language switching		8 languages: Hindi, Tamil, Telugu, French & more
No scheduling or reminder system		AI reads timetable & sends real email reminders
Generic answers for everyone		K-12 curriculum-aware, grade & board specific
No accessibility features		VI mode, Dyslexia mode, font & spacing controls

COMPLETE FEATURE SET — 20+ Features Built In

🎓 Core Learning

AI Teacher
4-step structured lessons

Auto Read Aloud
Every lesson read aloud

Practice Quiz
MCQ with AI feedback

Flashcards
Voice-controlled review

Exam Mode
Timed with score analysis

🤖 Agentic AI

Learning Agent
Analyses & adapts plan

Study Planner
Day-by-day AI schedule

Exam Prep Agent
Predicts exam questions

AI Coach
Motivation & daily goals

Doubt Solver
Instant doubt chat

♿ Accessibility

30+ Voice Cmds
Full voice app control

VI Mode
Screen-reader optimised

Dyslexia Mode
OpenDyslexic font

Font Customiser
Size, spacing controls

High Contrast
Low vision dark theme

⚙️ Smart Features

8 Languages
One-command switch

Timetable Upload
AI reads & sets alerts

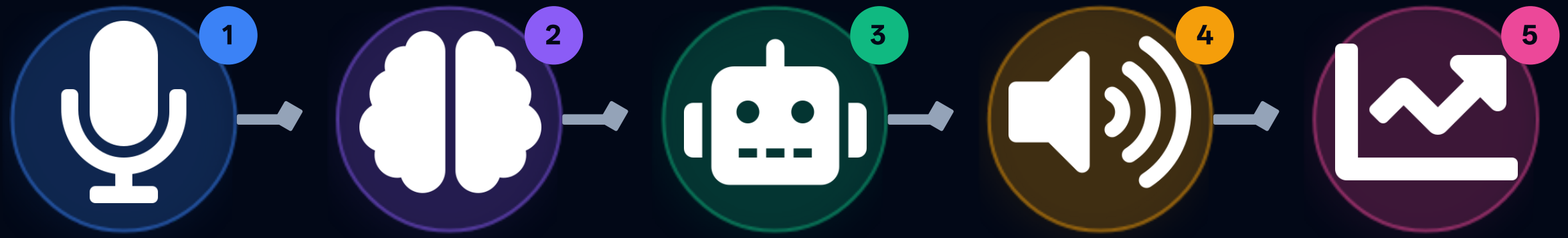
Email Reminders
Real emails via EmailJS

Progress Tracker
Streaks & weak topics

Spaced Repetition
SM-2 algorithm engine

HOW SARA AI WORKS

One student. One voice. Infinite learning.



Student
Speaks or types

Gemini AI
4 API keys process

SARA Acts
Teaches / Plans / Quizzes

Reads Aloud
In chosen language

Adapts
Tracks & improves

API KEY ARCHITECTURE — 40 Requests / Minute (4 Independent Keys)

Key 1
AI Teacher
Motivation Coach

Key 2
Practice Quiz
Lesson Generator

Gemini API

Key 3
Learning Agent
Exam Prep
Doubt Solver

Key 4
Study Planner
Flashcards
Timetable

ACCESSIBILITY & VOICE CONTROL

SARA works 100% by voice — no sight required.

VOICE COMMANDS (30+)

"teach me photosynthesis"	→ Opens AI Teacher & teaches topic
"generate quiz on history"	→ Instantly creates 10-question quiz
"flashcards on trigonometry"	→ Creates flip-card revision set
"answer B"	→ Selects quiz option — no keyboard!
"read question"	→ Re-reads current question aloud
"what is my score?"	→ Reads progress stats aloud
"change language to Hindi"	→ Switches entire app to Hindi
"set reminder for Maths exam"	→ Creates exam reminder by voice
"motivate me"	→ Opens Motivational Coach
"ask doubt: explain gravity"	→ Opens Doubt Solver with question

ACCESSIBILITY MODES

VI Mode

Large 22px font, wide line spacing 2.2, extra word spacing — optimised for low vision

Dyslexia Mode

OpenDyslexic font, 18px, line 2.0, letter spacing 1.5 — proven to aid reading difficulty

Full Read Aloud

Every heading, paragraph and result is automatically read in the student's language

Font Customiser

Font family, size (12–28px), line, letter and word spacing — all saved across sessions

8 Language UI

Voice, recognition and AI responses all switch together — Hindi, Tamil, Telugu & more

IMPACT & WHY SARA AI MATTERS

One project. One mission. Millions of students.

253M
VI People Worldwide

20+
AI Features Built In

30+
Voice Commands

8
Languages Supported

4
API Keys — Zero Rate Limit

Global Need

253 million visually impaired people lack accessible AI study tools. SARA fills that gap.

Completely Free

No subscriptions, no downloads, no server setup. Open any browser and start learning.

Inclusive by Design

Every single feature works by voice alone. Built from the ground up for VI students.

School Ready

Works for all K-12 grades, CBSE/ICSE/State boards, in 8 languages, anywhere in the world.

TECHNICAL ARCHITECTURE

TECH STACK

HTML5 + CSS3 + Vanilla JS

Single self-contained file. No frameworks, no build step, runs on any browser anywhere.

Google Gemini API

Auto-detects best model: gemini-2.5-flash → 2.0-flash → 1.5-flash

Web Speech API

Browser-native voice recognition & text-to-speech — zero third-party dependencies

EmailJS SDK

Real emails direct from browser — no backend or SMTP configuration needed

Rate Limit Strategy

4 keys across 4 Google projects = 40 combined requests/min, zero throttling

KEY INNOVATIONS

Robust JSON Parser

Custom array & object parsers handle incomplete AI responses — never crashes

Auto Model Detection

Tests gemini-2.5 → 2.0 → 1.5; always picks fastest available

Spaced Repetition

SM-2 algorithm schedules flashcard reviews at optimal memory intervals

Vision Timetable

Gemini vision reads handwritten timetable photos & extracts all exam dates

Creator Recognition

Detects creator's name → greets as 'Sir' with gold crown badge

AGENTIC AI — SARA'S BRAIN

SARA doesn't just answer — it thinks, plans and acts for you.

WHAT MAKES IT AGENTIC

Analyses Your History

Reads every quiz attempt, score and topic to understand exactly where you struggle

Builds a Study Plan

Generates a personalised day-by-day schedule prioritising your weakest areas

Predicts Exam Questions

Uses your syllabus + past patterns to predict the most likely questions

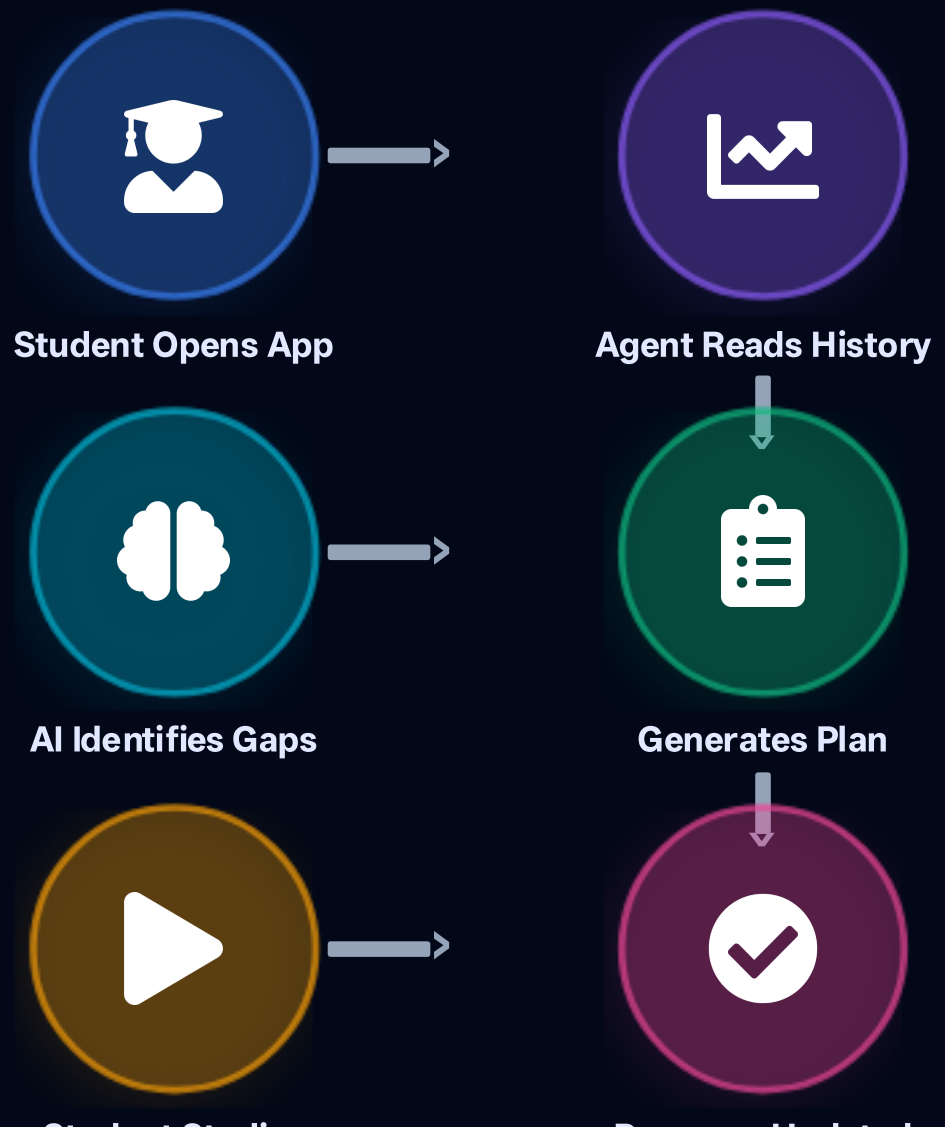
Spaced Repetition Engine

SM-2 algorithm reschedules flashcards based on how well you remembered them

Motivational Intelligence

Tracks mood, streak and scores to deliver the right encouragement at right time

AGENT WORKFLOW



8 LANGUAGES — GLOBAL REACH

One command switches everything — voice, AI responses and recognition.

Hindi
हिन्दी
600M speakers

Tamil
தமிழ்
80M speakers

Telugu
తెలుగు
95M speakers

English
English
1.5B speakers

French
Français
300M speakers

Spanish
Español
500M speakers

German
Deutsch
130M speakers

Italian
Italiano
85M speakers

🌍 Combined reach: 3.3+ Billion speakers • All 8 switch voice recognition + TTS + AI responses simultaneously • More languages coming soon

A DAY WITH SARA AI — STUDENT JOURNEY

From first wake-up to exam prep — SARA is with you every step.

7:00 AM

Morning Boost

SARA greets by name, reads the day's study goals, checks streak & sends motivational message

8:30 AM

Lesson Time

Student says 'teach me Newton's Laws' — SARA teaches in 4 steps, reads aloud in Hindi

11:00 AM

Practice Quiz

'Generate quiz on Newton's Laws' — 10 MCQs by voice, SARA gives instant audio feedback

2:00 PM

Flashcard Review

Spaced repetition schedules today's flashcards — student flips all 15 cards by voice

4:00 PM

Doubt Solving

'Ask doubt: explain electromagnetism' — SARA solves with example, quiz question & takeaway

6:00 PM

Study Planning

Learning Agent reviews today's performance & automatically adjusts tomorrow's study plan

8:00 PM

Exam Prep

Exam Prep Agent predicts likely Physics exam questions based on syllabus + history

9:00 PM

Reminder Set

SARA confirms: 'Physics exam in 3 days — reminder email sent to your inbox!' 📧

🔥 Every day SARA tracks your streak, score average & time studied — keeping students motivated & accountable

SMART REMINDERS — NEVER MISS AN EXAM

Upload a timetable photo. SARA does the rest.

HOW TIMETABLE REMINDERS WORK

1. **Upload Timetable**
   Photo or PDF of any timetable — even handwritten, even blurry

2. **Gemini Vision Reads**
   AI vision API reads & extracts all subjects, dates and times

3. **Reminders Created**
   All exam dates auto-added with customisable lead times (15min–1day)

4. **Voice Alert**
   At reminder time, SARA speaks aloud: 'Maths exam in 1 hour!'

5. **Email Sent**
   Real email via EmailJS with personalised message and study tip

SAMPLE REMINDER EMAIL

SARA AI — Exam Reminder 🎓

To: student@email.com
From: SARA AI Study Assistant
Subject: 📚 **Maths Exam Tomorrow!**

Hi Rahul,

Your Maths exam is scheduled for tomorrow, 10th June at 10:00 AM.

SARA's Tip: Review Chapter 7 — Trigonometry. You scored 65% on your last quiz on this topic. Spend 45 minutes on it tonight!

Your current streak: 🔥 12 days
Average score: 📊 78%

You've got this! — SARA 🤖

Powered by EmailJS • No backend server • Works from pure browser • Free tier: 200 emails/month

FUTURE ROADMAP — WHERE SARA IS HEADED

SARA v1 is just the beginning.

✅ SARA v1 — Available NOW

- 20+ AI features across learning, planning & accessibility
- 30+ voice commands for complete hands-free control
- 8 languages with full voice + AI switching
- Email reminders with AI timetable reading
- 4 Gemini API keys — 40 req/min combined
- Single HTML file — zero installation required

🚀 SARA v2 — Coming Next

- Mobile app (Android & iOS) with offline support
- Braille output integration for Braille displays
- Parent & teacher dashboard — track student progress
- School LMS integration (Google Classroom, Moodle)
- 15+ additional language support
- AI-generated personalised textbook summaries

⭐ SARA v3 — The Vision

- Real-time collaboration — study groups by voice
- ARCore audio AR — 3D spatial learning environments
- Medical & rehabilitation integration for VI students
- Government & NGO partnerships for free distribution
- Multi-modal: camera input for solving printed textbooks
- SARA certified curriculum for 50+ countries

Goal: Make SARA available to every visually impaired student on the planet — free, always.

Built with Responsible AI at its Core

We also ensured responsible AI design, focusing on **privacy, fairness, safety, and inclusion.**

No student data is stored on any server — all personalisation happens locally in the browser.

🔒 **PRIVACY**

⚖️ **FAIRNESS**

🛡️ **SAFETY**

♿ **INCLUSION**

ZERO DATA
stored on any server

100% LOCAL
All personalisation in browser

NO ACCOUNT
No tracking. No sign-up.

Safe for every student. Private by design. Open to all.

KEY LESSON LEARNT

Design with people, not just *for* them.

Our team learnt an important lesson — **real user feedback is essential** to building truly inclusive technology.

LISTEN FIRST
Talk to real users before building

KEEP ITERATING
Every problem became a feature

INCLUDE ALWAYS
Users shaped every decision

Tested with 3 visually impaired students • Special needs coordinator • 2 teachers • Laxman's feedback shaped every major change

The people you build for are the most important part of the process.

SARA AI

Smart Accessible Reader & Assistant

20+ Features	30+ Voice Commands	8 Languages	253M People	Free Forever

" Created with love for the ones who can make this world a beautiful place to live in "

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